Types of electronic waste

1. Discarded computer monitors
2. Motherboards
3. Cathode ray tubes (CRTs)
4. Printed circuit boards (PCBs)
5. Mobile phones and chargers
6. Compact disks
7. Headphones
8. Liquid-crystal display (LCD)/plasma televisions
9. White goods such as air-conditioners and refrigerators

Equipment used for e-waste recycling

1. Primary shredders
2. Secondary shredders
3. Eddy current separator
4. Vibratory feeder
5. Vibratory screen
6. Overband magnetic separator
7. Suspension magnet
8. Belt conveyor
9. Hammer mill
10. Drum separator
11. Electrostatic drum separator
12. Colour sorter
13. Metal detector
14. Magnetic pulley
15. Shaking table
16. Air separator
17. Electro-winning
18. Gyratory screen separator
19. Retorts

TABLE I
METHODS USED TO RECYCLE DIFFERENT TYPES OF E-WASTE COMPONENTS/PRODUCTS

e-waste components/products	Process used
Cathode ray tubes (used in TVs, computer monitors, ATM, video cameras, etc)	Breaking and removal of yoke, then dumping
Printed circuit board (a thin plate on which chips and other electronic components are placed)	Desoldering and removal of computer chips; open burning and acid baths to remove metals after chips are removed
Chips and other gold-plated components	Chemical stripping using nitric and hydrochloric acid and burning of chips
Plastics from printers, keyboards, monitors, etc	Shredding and low-temperature melting for reuse
Computer wires	Open burning and stripping to remove copper

TABLE II
METALS OBTAINED FROM DIFFERENT E-WASTE PRODUCTS/ COMPONENTS

e-waste products	Metals recovered after recycling
Nearly all electronic goods using more than a few watts of power (heat-sinks), electrolytic capacitors	Aluminium
Copper wire, printed circuit board tracks, component leads	Copper
Transistorised electronics (bipolar junction transistors) made in 1950s and 1960s	Germanium
Connector plating, primarily in computer equipment	Gold
Lithium-ion batteries	Lithium
Nickel-cadmium batteries	Nickel
Glass, transistors, ICs, printed circuit boards	Silicon
Solder, coatings on component leads	Tin
Plating for steel parts	Zinc

E-waste recycling process

E-waste recycling is highly labour-intensive and involves several steps starting from waste collection to final material processing (Fig. 1) in order to obtain the desired recycled products or raw materials for reuse.

Different types of e-waste materials and processes used for their management are laid out in Table I. The different valuable metals obtained after processing are given in Table II.

Business economics

e-waste business models include:

1. Franchisee-based operation: Waste management as a franchise unit of an already established e-waste management company

2. Self-owned operation: Waste management through purchase of waste from various sources

The franchisee model is suitable for investors who are looking for a business opportunity with low investment threshold. In this model, the franchisee unit could benefit from some of the parent company's infrastructure. Hence the capex and opex requirement is lower than for the self-owned business model. In the self-owned business model, the entrepreneur needs to bear the entire investment cost himself.

It's not possible to provide economic analysis of the franchisee model as it depends on the terms of agreement. However, typical requirements for starting a franchisee unit with a capacity to mobilise 60MT e-waste products per annum are:

1. Investment: ₹ 700,000-800,000

2. Area: 93-139sq.m (1000–1500sq.ft)

Under the self-owned model, a business unit capable of revenue generation to the tune of 6.0-7.5 million rupees per month (that is, processing of at least 500kg e-waste per day) are given below:

Capital expenses.

1. Land: 1858sq.m (20,000sq. ft) to start with

2. Plant and machinery: 3.5 million rupees

Operational expenses. Approximately 4 million rupees per month

Operational expenses include:

1. Human resource cost
2. Cost of scrap
3. Expenses on utilities such as electricity and water
4. Logistics expenses
5. Marketing expenses
6. Spares and consumables
7. Contingency expenses

Detailed business economics, including estimated value against the above-mentioned cost heads and other probable cost heads, along with profit analysis and breakeven point (BEP) calculation can be worked out based on the desired scale of operation.

Revenue calculation for recycling of 1-tonne personal computer (PC) e-waste.

1. PC e-waste: 1 tonne

2. Number of PCs: 38 (assuming that each PC weighs 28kg)

Scenario 1: Complete PGM recovery

Recovery of the precious platinum-group metals (PGMs) from PCBs is the main recycling revenue generator (refer Table III).

PCB constitutes almost 4 per cent